



OpenBook

E-Learning Course Progress Tracker

Project No. 199 | Full Stack Web Development

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Problem Statement

The challenge

A learner's progress is made up of completed lessons. Course completion must reflect every lesson, while each learner's progress remains private and course changes stay under the instructor's control.

What the system must enforce

1

Lesson-level tracking

Record completion for each learner and lesson.

2

All-lessons rule

A course is complete only when every current lesson is complete.

3

Access boundaries

Learners access their own progress; instructors manage courses they own.

4

Enrollment requirement

Progress actions are limited to learners enrolled in that course.

Our Solution: OpenBook

A full-stack course platform for learners and instructors.

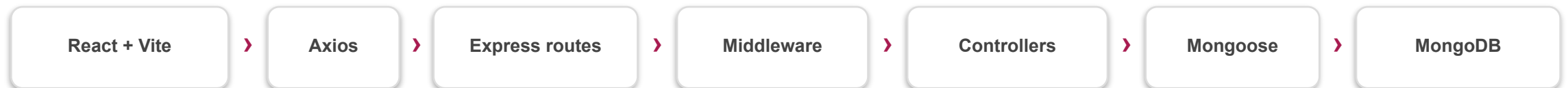
Learner workspace

- Browse courses and view lesson outlines.
- Enroll and read lesson content.
- Mark lessons complete and review personal progress.

Instructor workspace

- Create, edit, and delete owned courses.
- Maintain lesson titles, content, and order.
- Review enrollment and completion-rate analytics.

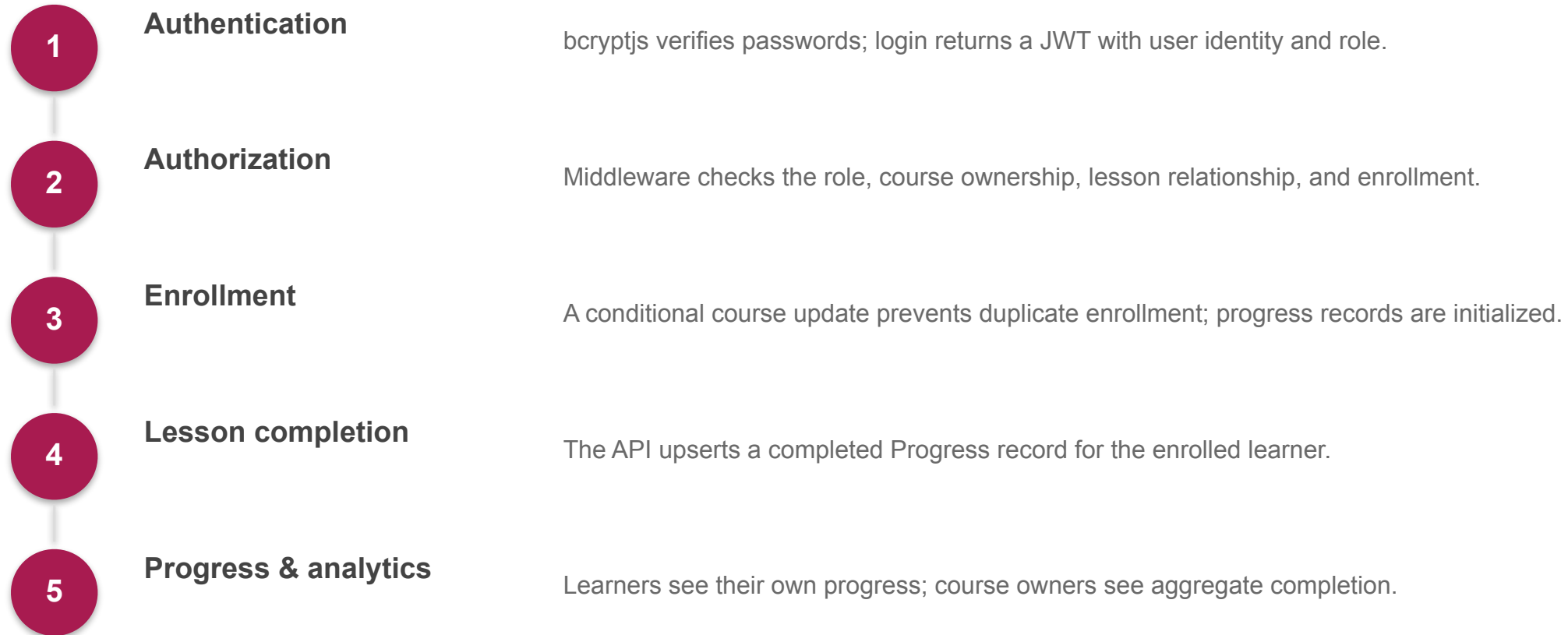
Backend request path



API responses use a consistent { message, data } envelope.



How OpenBook Works



Completion = completed current lessons ÷ total current lessons | Complete only when total > 0 and all lessons are done

Conclusion

60 / 60

Newman API requests passed

2

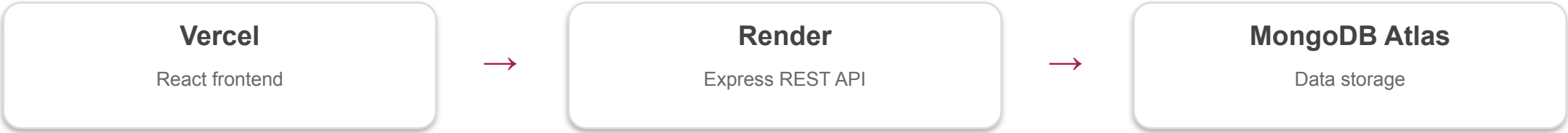
Account roles: learner and instructor

4

MongoDB collections: User, Course, Lesson, Progress

Course completion is calculated from lesson progress and enforced by the backend.

Deployment



GitHub Repository <https://github.com/sameershaikh-20/e-learning-course-progress-tracker>

